

```

astha@astha-VirtualBox:~$ ls
Desktop  devstack  Documents  Downloads  Music  Pictures  Public  snap  tacker  Templates  Videos
astha@astha-VirtualBox:~$ cd devstack/
astha@astha-VirtualBox:~/devstack$ ls
clean.sh      distribute_setup.py  extras.d  functions-common  HACKING.rst  LICENSE  openrc  roles  src  telenetry  tox.ini
CONTRIBUTING.rst  distribute_setup.py.1  files     FUTURE.rst      lib          local.conf  playbooks  run tests.sh  stackrc  tests  unstack.sh
data          doc                functions  Makefile         README.rst  Makefile  README.rst  samples  stack.sh  tools  vin_config.yaml
astha@astha-VirtualBox:~/devstack$

```

Figure 2: Folder snapshot

```

E: Package 'python-nose' has no installation candidate
+functions-common:apt_get:1
+./stack.sh:exit_trap:516
++./stack.sh:exit_trap:517
+./stack.sh:exit_trap:517
+./stack.sh:exit_trap:520
+./stack.sh:exit_trap:526
+./stack.sh:exit_trap:531
+./stack.sh:kill_spinner:426
+./stack.sh:exit_trap:533
+./stack.sh:exit_trap:534
Error on exit
+./stack.sh:exit_trap:536
+./stack.sh:exit_trap:537
+./stack.sh:exit_trap:539
+./stack.sh:exit_trap:542
/home/astha/devstack/tools/worldddump.py:22: DeprecationWarning: The distutils package is deprecated and slated for removal in Python 3.12. Use setuptools or check PEP 632 for potential alternatives
  from distutils import spawn
exit_trap
local r=100
jobs -p
jobs=
[[ -n '' ]]
[[ -f '' ]]
kill_spinner
[[ -z '' ]]
[[ 100 -ne 0 ]]
echo 'Error on exit'
type -p generate-subunit
generate-subunit 1667232758 5447 fail
[[ -z /opt/stack/logs ]]
/usr/bin/python3.10 /home/astha/devstack/tools/worldddump.py -d /opt/stack/logs

```

Figure 3: *Distutil lib* error

APIs, which allow applications and users to interact with them. These components communicate internally with each other to provide the necessary services to the applications and users. This modular architecture makes it possible to scale and customise the OpenStack infrastructure according to the needs of the users. The overview of OpenStack architecture is shown in Figure 1.

DevStack is an Ubuntu-based minimal installation of OpenStack (<https://docs.openstack.org/devstack/latest/>). It follows a certain template and installs all the components and services.

Step-by-step instructions for installing OpenStack and troubleshooting

The system configuration we have used for this experiment is as follows:

RAM: 8GB

Storage: 50GB

OS: Ubuntu 22.04

Kernel version: 5.15

OpenStack version: Yoga

DevStack version: 2023.1

Step 1: Install git as follows:

```
$ sudo apt-get install git
```

Step 2: Give the following command:

```
$ git clone https://opendev.org/openstack/devstack;
```

A folder with the name ‘devstack’ will be created.

Step 3: The output of ‘cd devstack’ will look like what’s shown in Figure 2.

Step 4: Create a new file named *local.conf* in the DevStack directory.

Step 5: In this file, copy the contents of the file present in the link <https://docs.openstack.org/tacker/latest/install/devstack.html>. The configuration file for OpenStack installation will contain several important parameters that are required for successful deployment, including:

- **Admin user and password:** These credentials will be used for authenticating and accessing the OpenStack dashboard.
- **MySQL password:** OpenStack uses MySQL as a database to store all its configuration and operational data. A strong MySQL password should be set to ensure security.
- **RabbitMQ password:** RabbitMQ is an open source message broker

software used by OpenStack services for communication. The password for RabbitMQ should be set securely.

- **Identity service (Keystone):** It configures the domain name, authentication method, token expiration time, etc.
 - **Compute service (Nova) configuration:** It contains details of hypervisor type, network configuration, and other compute-related settings.
 - **Network service (Neutron) configuration:** It holds the network types (VLAN, VXLAN, GRE), the network address ranges, and other network-related settings.
 - **Block storage service (Cinder) configuration:** This includes the storage backends, volume types, and other block storage-related settings.
- By configuring these parameters in the OpenStack installation configuration file, the OpenStack deployment will be tailored to meet the specific needs and requirements of the user. Then, we can start the installation to execute the *stack.sh* script.

If an error pops up as ‘cannot import spawn from distill’ then edit the